



Statewide Conference

Conference Theme: Teaching Post-COVID

April 12 - 13, 2024

Dear colleagues,

Greetings and a warm welcome to the VMATYC statewide conference! We are thankful to the Weyers Cave of the Blue Ridge Community College for hosting. We extend a special note of appreciation to Carlos Bolanos and Nathan Butler, whose hard work and dedication have been instrumental in organizing and setting up the conference, and to the esteemed speakers who will be presenting their insightful talks.

Additionally, we are grateful for the breakfast and snack provided by Cengage and Pearson, respectively.

Please keep in mind the following announcements:

- Your Vehicle should be parked outside of the Houff and S buildings. See the QR Code to the Right.
- For Campus WiFi, use BRCC guest and log in with your VCCS Login Credentials.
- Consider joining VMATYC as a member! Join now to receive membership through December 2024. Visit <http://vmatyc.org/join-vmatyc/>
- To stay updated with the most recent VMATYC developments, make sure to visit our website at <https://vmatyc.org/> and read our newsletter.



Sharon Schoolcraft
Laurel Ridge Community College
VMATYC Northern Region Vice President

Michael Filsinger
Virginia Peninsula Community College
VMATYC President

Agenda

Friday

Opening: Registration [S Building]

1:00pm – 2:00pm

Session One: 2:00pm – 2:50pm

Session Two: 3:00pm – 3:50pm

Session Three: 4:00pm – 4:50pm

Dinner Banquet: Keynote Speaker [Plecker 126 A&B] 5:30pm – 6:30pm

Saturday

Session Four: 9:00am – 9:50am

Session Five: 10:00am – 10:50am

Session Six: 11:00am – 11:50am

Boxed Lunch: Business Meeting [Plecker 126 A&B]

12:00pm – 1:00pm

Presentation Descriptions

Keynote Speaker: Dr. Patrice Waller “Reimagining Mathematics Education: Addressing Educational Challenges in a “Post-COVID” World”[Plecker 126 A&B]

Session One (2:00pm - 2:50pm)

The Prime Number Theorem [S108] *Chamath Hettiarachchi*, Northern Virginia Community College Annandale Campus

Speaker will discuss the proof of The Prime Number Theorem.

Project-Based Learning: Applied Projects for Student Engagement [S111 PC Lab] *Samuel Chukwuemeka*, Blue Ridge Community College

Attend this interactive workshop to discuss (share and/or learn) projects that connect the knowledge acquired in the classroom to applications outside the classroom, while keeping our students engaged in the learning process. We shall discuss applied projects in several undergraduate mathematics courses.

Session Two (3:00pm - 3:50pm)

Artificial Intelligence: Where Are We Going, Where Have We Been? Understanding the AI Underpinning the ALEKS Adaptive Engine [S108]
Jason Kanevsky & Courtney Brandt, McGraw Hill

Knowledge Space Theory is a theory developed in the 1980s to describe the artificial intelligence that underpins the ALEKS adaptive learning engine. In this talk, we will introduce Knowledge Space Theory and uncover how ALEKS uses large datasets to make predictions about what students are ready to learn. We will also discuss how the development of neural networks has improved the adaptive engine in ALEKS, allowing the system to make more accurate predictions about ready to learn topics.

Bridging the Knowledge Gap and Encouraging Equity with Atal’s Dynamic Remediation [S109]
Emily Garvin & Grace Bush, Wiley

The mission of Wiley’s affordable, accessible, and adaptive learning platform, Alta, is to bridge knowledge gaps and put achievement within reach for all students. In this presentation, we will explore how the power of adaptive learning can bridge prerequisite knowledge gaps and address state curriculum requirements without asking instructors to stretch themselves even further than they already do. Integral to this discussion will be an exploration of the qualitative and quantitative data about the success of Alta and adaptive learning as an equitable solution for students across all levels of math in higher education.

Enhancing Line Equation Understanding: Incorporating the x-Intercept [S119] *Shahrokh Ahmadi*, Northern Virginia Community College

This talk presents a critical addition to the understanding of linear equations by introducing the equation of a line in terms of its x-intercept. While the traditional form, $y = mx + b$, with the y-intercept 'b' is well-known, this paper sheds light on the less familiar form, $y = m(x - a)$, where 'a' represents the x-intercept. Despite its absence from many algebra textbooks, this form is fundamental in numerous practical scenarios. This paper discusses the derivation of this equation, its significance, and provides a practical application to demonstrate its utility.

Session Three (4:00pm - 4:50pm)

Research on NSF Direct Enrollment: Initial Findings and Next Steps [S108] *Zack Beamer*, Piedmont Virginia Community College

All VCCS schools are now using MDE prerequisite and corequisite courses as the developmental on-ramp to gateway mathematics, using GPA data in lieu of high stakes placement. Thanks to an NSF grant, we are finally getting the opportunity to systematically study the implementation, impact, and instruction of the courses in the Direct Enrollment reforms across the system. This talk will overview the scope of the research project and the current state of research. It will also present initial quantitative findings on the impact of placement into corequisite support on student success outcomes at one community college within the system. Your feedback is highly valued as it will help inform the next steps of research and contribute to the quality of data gathered and enhance the conclusions the research team is able to draw.

Imagining an OER Future for Math Education in the VCCS [S109] *Cody Fouts*, Laurel Ridge Community College

In this presentation, I will share how I have used OER (Open Education Resource) materials in my classes throughout my teaching career. I will also share some of the resources available to instructors who are interested in dipping their toes into the wonderful world of OER. The ultimate goal of the talk is to begin to build a Math OER Community in Virginia to encourage the sharing of resources and materials among instructors.

Using Python in Differential Equations [S117] *Irina Liskey*, Piedmont Virginia Community College

This presentation explores applying Python in modeling differential equations. We delve into the fundamentals of Python and its powerful libraries such as NumPy, SciPy, and SymPy for solving different equations. Through practical examples and code demonstrations, attendees will gain insights into implementing and using these methods for their classes.

Session Four (9:00am - 9:50am)

Math Talks Across the Grades: Accessible and Engaging for Everyone [S108] *Keith Nabb*, Piedmont Virginia Community College

Experience how the Number Talk idea can extend to almost any mathematical content, K-14. Examples will be drawn from Arithmetic, Problem Solving, Patterns, Algebra, Precalculus/Trigonometry, Calculus, and beyond. This session will share examples of student-led Math Talks and insights gained from implementation.

A Brief History of the Philosophy of Mathematics [S109] *Johannes Memling*, Northern Virginia Community College, Annandale Campus

We will be discussing the history of the philosophy of mathematics to connect it to its presence and possible applications in the future of mathematics and the advancing technologies associated with it.

AI Roundtable [S119] *Wendi Dass & Zack Beamer*, Piedmont Virginia Community College

Concerned about the use of AI in your classes? Join your colleagues for a discussion on the capabilities and limitations of generative AI in the solution of mathematics problems. The roundtable facilitators will also offer advice from their own experiences and provide tips for writing assessments that require your students (not ChatGPT) to do the work.

Session Five (10:00am - 10:50am)

Google Sites: Design Your Professional Classroom [S108] *Samuel Chukwuemeka*, Blue Ridge Community College

Would it not be great to have your contents (notes, lesson plans, links to videos, assignments, tests, and other resources) in one place: your website? How can we create teaching and learning opportunities beyond the physical classroom and/or the course management system? (2 hours)

Creating Shareable Canvas Templates with Extensive Videos, PowerPoints, and Fine-tuned Assignments to Improve Student Success and Reduce Financial Burdens with Open Educational Resources (OER) [S117] *Amanda Maule & Amy Schmidt*, Virginia Peninsula Community College

Come learn tips for building and sharing Canvas course templates with organized, resource-rich lessons and projects across face-to-face and online courses, lowering costs to students while increasing satisfaction. Due to the abrupt switch to online education during COVID, VPCC faculty began increased collaboration by contributing resources such as video lectures, PowerPoints, and well-crafted assignments and projects, to create Canvas master templates which are easily shareable among faculty. These methods fostered success among adjunct faculty and those new to teaching a course, which culminated in student success during COVID and afterward. As a result, we have observed increased pass rates and satisfaction among students and faculty in MTH 154 Quantitative Reasoning, MTH 155 Statistical Reasoning, and MTH 245 Elementary Statistics, while allowing creation of a MTH 155 Statistical Reasoning course using OER (open educational resources). This OER version of MTH 155 is now taught across our college and is soon to be replicated in high school dual enrollment courses.

Using OBS (Open Broadcaster Software) in Synchronous teaching modality [S111 PC Lab] *Sergio Maria-Fagundez*, Virginia Peninsula Community College, Hampton Campus

Discover the power of OBS, a free software tool that can greatly enhance your synchronous classes.

Session Six (11:00am - 11:50am)

The Hidden Beauty of Pi [S109] *Emmanuel Addo*, Northern Virginia Community College

It is an undisputed fact that pi has received unlimited praises for centuries. The mystery of this well-known constant is the fact that its exactness lies in inexactness. The use of the symbol as well as the significance of the pi cannot be over emphasized. In this presentation, I intend to give a brief history of pi including the analysis and the numerical calculations. Discuss certain events associated with pi and the beauty in the pattern of its digits. Give examples of formulas containing pi in the world of science, mathematics, and beyond, and then proceed to explain how pi has led to resolving some mathematical problems in the early 19th century. Stating some roles that pi plays in measurements and its practical contribution to computers. I then conclude with open questions about pi.

Epistemology of Distance [S119] *Jorge Menacho*, Northern Virginia Community College, Annandale Campus

In this presentation, I plan to examine the concept of distance. We will explore some ontological, phenomenological, and mathematical questions about the nature of distance and compare the multiple ideas that exist about this concept. The main goal of this inquiry is to approach some sort of Platonic form or essence of what distance is in itself, such that we can connect our commonsense or intuitive notions of distance that we get from experience to the more abstract notions of it that we find in mathematics and physics.

Design of 7 x 7 MTH 245 Course – Statistics [S109] *Ebenezer Emos*, Brightpoint Community College

Discuss the design processes such as a seminar on developing the 7 x 7 courses, developing the course mapping, developing the pacing guidelines, developing the course in Pearson, course implementation and feedback.